

Online Bookstore Database Management System

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Description

This project implements a robust and normalized database system for an online bookstore using MySQL. The system covers all core operations such as managing customers, books, orders, payments, and reviews. Additional features include triggers, views, and scheduled automation using events for real-time and analytical operations.

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Objectives

- Design a normalized relational database schema for an online bookstore.
- Implement core functionalities including customer management, order handling, and payments.
- Use views, triggers, and events for automation, reporting, and data integrity.
- Enable future extensibility through a clean schema and modular logic.

Core Entities

- Customer
- Book
- Author
- Publisher
- Category
- Orders
- OrderItem
- Payment
- Review

Key Features Implemented

- Automatic stock update using triggers upon new orders.
- Notification logs for low-stock and pending orders via event scheduling.
- Price change logging and admin tracking.
- Monthly and daily report generation through events.
- Views for customer summaries and category-wise revenue.

Reports & Queries Highlights

- Retrieve Books in "Fiction" Category That Are In Stock
- Top 3 Best-Selling Books of the Year
- Books with No Purchases
- Average rating of books with at least 10 reviews
- Orders that include at least one book from the 'Science & Technology' category
- Pending orders with customer contact

Techs N Tools Used

- MySQL 8.x
- SQL (DDL, DML, Triggers, Views, Events)
- MySQL Workbench (for schema visualization)

Project File Structure

- `readme.pdf`– This very file. Info about the project
- `schema-build.sql` – All CREATE TABLE statements and constraints
- `sample-data-strips.sql` – Sample data for books, customers, orders, etc.
- `queries.sql` – Analytical and operational SQL queries
- `stored-procedures.sql` – Triggers and procedures with custom delimiters
- `automated-operations.sql` – Scheduled events and automation
- `virtual-tables.sql` – SQL views for summaries and reports
- `presentation.pptx` – Presentation of the project

Key Learnings

- Practical schema design through normalization
- Writing and debugging real-world triggers and events
- Creating modular ERDs for scalable system planning
- Efficient SQL querying across multiple related entities
- Managing relational integrity and automation logic

Future Scope

- Add email-based alert integration for inventory and order notifications
- Build a user-facing frontend with a shopping cart and review system
- Connect the system with third-party APIs for payment and delivery tracking
- Implement loyalty programs and personalized recommendations
- A GUI for intuitive interaction

Bibliography

- **Database System Concepts** (7th Edition) by Silberschatz, Korth & Sudarshan – for conceptual understanding
- **MySQL 8.x Official App**– for syntax, examples, and DBMS features
- **ChatGPT by OpenAI** – for generating sample data strips and query assistance